

Sufficiency

Math 742

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Day 1

Introduction

The overarching goal of statistics is to take a sample of data from some population and use the information in the sample to make inferences about the population. More specifically, if we have a sample $X_1, X_2, \dots, X_n$ we would like to make inferences regarding some population parameter θ .

Suppose we have a large sample $x_1, x_2, \dots$. Because of the size of the sample, it might be hard to interpret a large list of numbers. Instead we use something called a **statistic**. A **statistic** is any deterministic function of the data that yields a summary. Different samples may produce the same statistic, which means the statistic loses information compared to the full data.

Typical statistics we can calculate are: sample mean ($\bar{x}$), sample variance (s^2), the max value in the sample, the min value in the sample, etc.

Let $\mathbf{X} = X_1, X_2, \dots, X_n$ and $\mathbf{x} = x_1, x_2, \dots, x_n$ denotes the sample values. Any statistic $T(\mathbf{X})$ defines a form of data reduction or data summary. Notice that if we only want to report the statistic rather than the entire sample ($\mathbf{x}$), we may have different samples that yield the same statistic.

For example, $\mathbf{x} = (1, 4, 5, 6, 9)$, $\bar{x} = 5$ and $\mathbf{y} = (1, 2, 5, 7, 10)$, $\bar{y} = 5$. Notice that $T(\mathbf{x})$ doesn't report the actual values, just the mean value.

There are multiple methods of data reduction, the topic for this lecture is the **Sufficiency Principle**.

Definition 1: A statistic $T(\mathbf{X})$ is a *sufficient statistic* for θ if the conditional distribution of the sample $\mathbf{X}$ given the value of $T(\mathbf{X})$ does not depend on θ .

Basically, once you know $T(\mathbf{X})$ the full sample $\mathbf{X}$ would not change your inference about θ .

Suppose $X_1, X_2, \dots, X_n \sim \text{Poisson}(\lambda)$. and consider the statistic:

$$T(\mathbf{X}) = X_1 + X_2 + \dots + X_n = \sum_{i=1}^n X_i$$

- The conditional distribution of the individual X_i 's given the sum $T(\mathbf{X})$ does not depend on λ .
- Therefore, $T(\mathbf{X})$ is sufficient for λ .
- If you know $T(\mathbf{X})$, you don't need the full dataset to calculate λ .

Here's an analogy. Suppose the whole book is $\mathbf{X}$ and θ is a secret message hidden in the book. If $T(\mathbf{x})$ is one paragraph that summarizes the secret message (i.e. θ). Once the paragraph is known, the entire book contains no additional information about the secret (i.e. θ).

Ratio Method: The ratio method is an easy way to show that a statistic $T(\mathbf{X})$ is sufficient for θ .

By **definition 1**, the conditional distribution of $\mathbf{X}|T(\mathbf{X})$ doesn't depend on θ .

$$f_{\mathbf{X}|T(\mathbf{X})}(x|t) = \frac{f_{\mathbf{X}}(x)}{f_{T(\mathbf{X})}(t)} \mathbf{1}\{T(\mathbf{x}) = t\}$$

The indicator function enforces the restriction that only sample points with $T(x) = t$ have positive conditional probability.

Note: This follows because X is discrete; for continuous variables, the same reasoning applies with pdfs.

If this ratio doesn't include θ , then $T(\mathbf{X})$ is a sufficient statistic.

We take the ratio because conditional probability is "joint divided by marginal." If that conditional doesn't depend on θ , then knowing $T(\mathbf{X})$ makes the distribution of the sample independent of θ , so T must contain all the information about θ .

Example: Let $X_1, X_2, \dots, X_n$ be a sample of independent Bernoulli random variables (i.e. 1's and 0's). Let $T(\mathbf{X}) = X_1 + X_2 + \dots + X_n$. Show that $T(\mathbf{X})$ is a sufficient statistic for p , the probability of success.

The PMF for the Bernoulli distribution is:

$$p(x_i) = p^{x_i}(1-p)^{1-x_i}, x_i = 0, 1$$

The joint PMF of the sample $\mathbf{X}$ is,

$$\begin{aligned} p(\mathbf{x}|p) &= \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i} \\ &= p^{\sum x_i} (1-p)^{\sum (1-x_i)} \end{aligned}$$

Note: The Joint PMF being the product of the individual PMFs only works because the random variables are independent.

Let $t = \sum x_i$, then

$$p(\mathbf{x}|p) = p^t(1-p)^{n-t}$$

The marginal PMF of $T(\mathbf{X})$ is a binomial distribution, so its PMF is equal to,

$$q(T(\mathbf{x})|p) = \binom{n}{t} p^t(1-p)^{n-t}$$

Take the ratio,

$$\frac{p(\mathbf{x}|p)}{q(T(\mathbf{x})|p)} = \frac{p^t(1-p)^{n-t}}{\binom{n}{t} p^t(1-p)^{n-t}} = \frac{1}{\binom{n}{t}}, \text{ where } t = \sum x_i$$

This ratio doesn't contain p , so the statistic $T(\mathbf{X})$ is sufficient for p .

Factorization Theorem: Let $f(\mathbf{x}|\theta)$ denote the joint PDF or PMF for a sample $\mathbf{X}$. Suppose $T(\mathbf{X})$ is a sufficient statistic iff there exists functions $g(t|\theta)$ and $h(\mathbf{x})$ such that all sample points $\mathbf{x}$ and parameter points θ ,

$$f(\mathbf{x}|\theta) = g(T(\mathbf{X})|\theta)h(\mathbf{x}), \quad \text{for all } \mathbf{x}, \theta$$

The idea is that Sufficiency means all the θ information is squeezed into a single function of the data.

Example: For the exponential distribution, show that $T = \sum X_i$ is sufficient.

Let $X_1, X_2, \dots, X_n$ be a iid sample from an exponential distribution with rate parameter λ .

For any one X_i the PDF is:

$$f_X(x|\lambda) = \begin{cases} \lambda e^{-\lambda x} : & x \geq 0 \\ 0 : & \text{otherwise} \end{cases}$$

The joint density is just the product of the densities for each X_i .

$$\begin{aligned} f(\mathbf{x}|\lambda) &= \prod_{i=1}^n f_X(x_i|\lambda) = \prod_{i=1}^n (\lambda e^{-\lambda x_i}) \\ &= \lambda^n e^{-\lambda \sum x_i} \end{aligned}$$

If T is the sum of the X_i 's, then

$$f(\mathbf{x}|\lambda) = \lambda^n e^{-\lambda T(\mathbf{x})}$$

Notice that the right hand side of the equation is $g(T(\mathbf{x})|\lambda)$. If we make $h(\mathbf{x}) = 1, x_i \geq 0$, then we have

$$f(\mathbf{x}|\lambda) = g(T(\mathbf{x})|\lambda)h(\mathbf{x})$$

So by the factorization theorem, $T(\mathbf{x})$ is sufficient for λ .

Example: Let $X_1, X_2, \dots, X_n$ be iid sample from a discrete uniform distribution on $1, \dots, \theta$. That is, the unknown parameter θ is a positive and the PMF of X_i is,

$$f_X(x|\theta) = \begin{cases} \frac{1}{\theta} : & x = 1, 2, \dots, \theta \\ 0 : & \text{otherwise} \end{cases}$$

Show that the maximum order statistic $T(\mathbf{x}) = \max(x_1, x_2, \dots, x_n) = X_{(n)}$ is sufficient for all $\mathbf{x}$ and θ .

The joint PMF is:

$$f_X(\mathbf{x}|\theta) = \begin{cases} \theta^{-n} : & x_i \in \{1, \dots, \theta\} \text{ for } i = 1, 2, \dots, n \\ 0 : & \text{otherwise} \end{cases}$$

Note: This will be the joint density only when $x_i \in \{1, \dots, \theta\} \iff X_{(n)} \leq \theta$. The restriction, $x_i \in \{1, \dots, \theta\}$ for $i = 1, 2, \dots, n$ can be re-expressed as $x_i \in \{1, 2, \dots\}$ for $i = 1, 2, \dots, n$ where $X_{(n)} \leq \theta$.

Define $T(\mathbf{X})$ be the $X_{(n)}$

Let,

$$h(x) = \begin{cases} 1 : & x_i \in \{1, 2, \dots\} \text{ for } i = 1, 2, \dots, n \\ 0 : & \text{otherwise} \end{cases}$$

and,

$$g(t|\theta) = \begin{cases} \theta^{-n} : & t \leq \theta \\ 0 : & \text{otherwise} \end{cases}$$

This makes $f_X(x|\theta) = g(t|\theta)h(x)$ and so by the factorization theorem, $T(\mathbf{x}) = X_{(n)}$ is sufficient for all $\mathbf{x}$ and θ .

Factorization Theorem.

Let $f(\mathbf{x} \mid \theta)$ denote the joint PMF of a discrete sample $\mathbf{X}$. A statistic $T(\mathbf{X})$ is sufficient for θ if and only if there exist functions $g(t \mid \theta)$ and $h(\mathbf{x})$ such that

$$f(\mathbf{x} \mid \theta) = g(T(\mathbf{x}) \mid \theta) h(\mathbf{x}) \quad \text{for all } \mathbf{x}, \theta.$$

Proof ($\Rightarrow$ direction):

Assume $T(\mathbf{X})$ is a sufficient statistic for θ .

We work in the discrete case.

Define

$$g(t \mid \theta) = \mathbb{P}_\theta(T(\mathbf{X}) = t), \quad h(\mathbf{x}) = \mathbb{P}(\mathbf{X} = \mathbf{x} \mid T(\mathbf{X}) = T(\mathbf{x})).$$

By the definition of sufficiency, the conditional distribution $\mathbb{P}_\theta(\mathbf{X} = \mathbf{x} \mid T(\mathbf{X}) = t)$ does not depend on θ , and therefore $h(\mathbf{x})$ is free of θ .

Now observe that

$$f(\mathbf{x} \mid \theta) = \mathbb{P}_\theta(\mathbf{X} = \mathbf{x}) = \mathbb{P}_\theta(\mathbf{X} = \mathbf{x}, T(\mathbf{X}) = T(\mathbf{x})).$$

Since T is a deterministic function of $\mathbf{X}$, the event $\{\mathbf{X} = \mathbf{x}\}$ implies the event $\{T(\mathbf{X}) = T(\mathbf{x})\}$. Thus,

$$\{\mathbf{X} = \mathbf{x}\} \subseteq \{T(\mathbf{X}) = T(\mathbf{x})\},$$

and therefore

$$\{\mathbf{X} = \mathbf{x}\} = \{\mathbf{X} = \mathbf{x}, T(\mathbf{X}) = T(\mathbf{x})\}.$$

Using the definition of conditional probability (and assuming $\mathbb{P}_\theta(T(\mathbf{X}) = T(\mathbf{x})) > 0$),

$$\begin{aligned} & \mathbb{P}_\theta(\mathbf{X} = \mathbf{x}) \\ &= \mathbb{P}_\theta(\mathbf{X} = \mathbf{x}, T(\mathbf{X}) = T(\mathbf{x})) \\ &= \mathbb{P}_\theta(T(\mathbf{X}) = T(\mathbf{x})) \mathbb{P}_\theta(\mathbf{X} = \mathbf{x} \mid T(\mathbf{X}) = T(\mathbf{x})). \end{aligned}$$

That is, the probability of observing $\mathbf{x}$ can be decomposed into the probability of observing its statistic $T(\mathbf{x})$ and the conditional probability of $\mathbf{x}$ given that statistic.

Substituting the definitions of g and h gives

$$f(\mathbf{x} \mid \theta) = g(T(\mathbf{x}) \mid \theta) h(\mathbf{x}),$$

which establishes the desired factorization.

For iff we have to show the other direction.

($\Leftarrow$ direction):

Assume that the joint PMF admits the factorization

$$f(\mathbf{x} \mid \theta) = g(T(\mathbf{x}) \mid \theta) h(\mathbf{x})$$

for all $\mathbf{x}$ and θ .

Fix a value t in the range of T . Then

$$\mathbb{P}_\theta(T(\mathbf{X}) = t) = \sum_{\mathbf{y}: T(\mathbf{y})=t} f(\mathbf{y} \mid \theta) = \sum_{\mathbf{y}: T(\mathbf{y})=t} g(t \mid \theta) h(\mathbf{y}).$$

Since $g(t \mid \theta)$ does not depend on $\mathbf{y}$, it factors out:

$$\mathbb{P}_\theta(T(\mathbf{X}) = t) = g(t \mid \theta) \sum_{\mathbf{y}: T(\mathbf{y})=t} h(\mathbf{y}).$$

Now consider the conditional distribution of $\mathbf{X}$ given $T(\mathbf{X}) = T(\mathbf{x})$:

$$\mathbb{P}_\theta(\mathbf{X} = \mathbf{x} \mid T(\mathbf{X}) = T(\mathbf{x})) = \frac{f(\mathbf{x} \mid \theta)}{\mathbb{P}_\theta(T(\mathbf{X}) = T(\mathbf{x}))}.$$

Substituting the expressions above,

$$\mathbb{P}_\theta(\mathbf{X} = \mathbf{x} \mid T(\mathbf{X}) = T(\mathbf{x})) = \frac{g(T(\mathbf{x}) \mid \theta) h(\mathbf{x})}{g(T(\mathbf{x}) \mid \theta) \sum_{\mathbf{y}: T(\mathbf{y})=T(\mathbf{x})} h(\mathbf{y})} = \frac{h(\mathbf{x})}{\sum_{\mathbf{y}: T(\mathbf{y})=T(\mathbf{x})} h(\mathbf{y})}.$$

This conditional probability does not depend on θ . Therefore, by definition, $T(\mathbf{X})$ is a sufficient statistic for θ .

Day 2

First we will express the Poisson model in canonical form for one observation to identify the basic sufficient statistic,

$$\text{Conical Form: } p(x|\eta) = h(x) \exp(\eta T(X) - A(\eta))$$

$$\text{Poisson PDF: } \mathbb{P}(X = x|\lambda) = \frac{1}{x!} \exp(\eta x - e^\eta)$$

$$h(x) = \frac{1}{x!}$$

$$T(x) = x$$

$$A(\eta) = e^\eta$$

For this case, (i.e. one sample) our sufficient statistic is $T(x) = x$.

When we observe an i.i.d. sample, the joint likelihood accumulates information additively, so the sufficient statistic for the sample is $T(x) = \sum_{i=1}^n x_i$.

Review: Sufficiency via the Factorization Theorem

Recall the *Factorization Theorem*:

A statistic $T(\mathbf{X})$ is sufficient for a parameter θ if and only if the joint pmf (or pdf) can be written in the form

$$f_{\mathbf{X}}(\mathbf{x} \mid \theta) = g(T(\mathbf{x}) \mid \theta) h(\mathbf{x}),$$

where g depends on $\mathbf{x}$ only through $T(\mathbf{x})$ and h does not depend on θ .

We apply this result to the Poisson model.

Example: Poisson Distribution.

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$. The joint pmf of $\mathbf{X} = (X_1, \dots, X_n)$ is

$$f_{\mathbf{X}}(\mathbf{x} \mid \lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i} \frac{1}{\prod_{i=1}^n x_i!}.$$

Rewriting this expression,

$$f_{\mathbf{X}}(\mathbf{x} \mid \lambda) = \underbrace{e^{-n\lambda} \lambda^{\sum x_i}}_{g(T(\mathbf{x}) \mid \lambda)} \underbrace{\frac{1}{\prod x_i!}}_{h(\mathbf{x})},$$

where

$$T(\mathbf{X}) = \sum_{i=1}^n X_i.$$

All dependence on λ occurs through $T(\mathbf{x})$. Therefore, by the Factorization Theorem,

$$T(\mathbf{X}) = \sum_{i=1}^n X_i$$

is a sufficient statistic for λ .

Connection to the Ratio Method.

The ratio method formalizes this same idea: if two samples $\mathbf{x}$ and $\mathbf{y}$ satisfy $T(\mathbf{x}) = T(\mathbf{y})$, then their likelihood ratio

$$\frac{f_{\mathbf{X}}(\mathbf{x} \mid \lambda)}{f_{\mathbf{X}}(\mathbf{y} \mid \lambda)}$$

does not depend on λ . Thus, the ratio method provides an alternative way to verify sufficiency without explicitly computing a marginal distribution.

The conical (or canonical) form of the exponential family is a special case of a more general formulation. It is extremely useful for building intuition, since the sufficient statistic is immediately visible. However, to prove sufficiency in full generality—especially for multi-parameter models—we need a more flexible representation of the exponential family. The form introduced by Casella and Berger allows for multiple statistics and parameters, and makes the connection to the Neyman–Fisher Factorization Theorem precise. We now switch to that more general form for the proof.

The conical form corresponds to the special case $k = 1$, with $c(\theta) = \exp(-A(\eta))$.

Theorem: Let $X_1, X_2, \dots, X_n$ be iid observations from a PDF or PMF, $f(x \mid \theta)$ that belongs to an exponential family given by,

$$f(x \mid \theta) = h(x)c(\theta) \exp \left(\sum_{i=1}^k w_i(\theta) t_i(x) \right)$$

where $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_d)$ ($d \leq k$), then

$$T(\mathbf{X}) = \left(\sum_{j=1}^n t_1(X_j), \sum_{j=1}^n t_2(X_j), \dots, \sum_{j=1}^n t_k(X_j) \right)$$

is a sufficient statistic for $\boldsymbol{\theta}$.

Proof: Let $X_1, \dots, X_n$ be i.i.d. observations from a density or mass function belonging to a k -parameter exponential family,

$$f(x | \boldsymbol{\theta}) = h(x) c(\boldsymbol{\theta}) \exp \left(\sum_{i=1}^k w_i(\boldsymbol{\theta}) t_i(x) \right).$$

Since $X_1, \dots, X_n$ are i.i.d., their joint density is

$$f(\mathbf{x} | \boldsymbol{\theta}) = \prod_{j=1}^n f(x_j | \boldsymbol{\theta}) = \left(\prod_{j=1}^n h(x_j) \right) c(\boldsymbol{\theta})^n \exp \left(\sum_{j=1}^n \sum_{i=1}^k w_i(\boldsymbol{\theta}) t_i(x_j) \right).$$

Interchanging the order of summation gives

$$\sum_{j=1}^n \sum_{i=1}^k w_i(\boldsymbol{\theta}) t_i(x_j) = \sum_{i=1}^k w_i(\boldsymbol{\theta}) \left(\sum_{j=1}^n t_i(x_j) \right).$$

Define, for $i = 1, \dots, k$,

$$T_i(\mathbf{x}) = \sum_{j=1}^n t_i(x_j), \quad T(\mathbf{x}) = (T_1(\mathbf{x}), \dots, T_k(\mathbf{x})).$$

Then we can write

$$f(\mathbf{x} | \boldsymbol{\theta}) = \underbrace{\left(\prod_{j=1}^n h(x_j) \right)}_{h(\mathbf{x})} \underbrace{c(\boldsymbol{\theta})^n \exp \left(\sum_{i=1}^k w_i(\boldsymbol{\theta}) T_i(\mathbf{x}) \right)}_{g(T(\mathbf{x}) | \boldsymbol{\theta})}.$$

This is the Neyman–Fisher factorization $f(\mathbf{x} | \boldsymbol{\theta}) = g(T(\mathbf{x}) | \boldsymbol{\theta}) h(\mathbf{x})$, where $h(\mathbf{x})$ is free of $\boldsymbol{\theta}$. Therefore $T(\mathbf{X})$ is sufficient for $\boldsymbol{\theta}$.

The importance of this theorem is hopefully obvious. Recall that using the Conical form (i.e. Wasserman) of the exponential family, the Poisson distribution can be put in the form of the exponential family.

Normal Distribution and Sufficiency (Two Cases)

Case 1: σ^2 known, μ unknown

Let

$$X_1, \dots, X_n \sim N(\mu, \sigma^2), \quad \sigma^2 \text{ known.}$$

The joint density is

$$f(\mathbf{x} \mid \mu) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right).$$

Expand the quadratic:

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2.$$

Substituting,

$$f(\mathbf{x} \mid \mu) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2 + \frac{\mu}{\sigma^2} \sum_{i=1}^n x_i - \frac{n\mu^2}{2\sigma^2}\right).$$

This can be written in exponential-family (conical) form:

$$f(\mathbf{x} \mid \mu) = h(\mathbf{x}) c(\mu) \exp(w(\mu) T(\mathbf{x})),$$

where

$$T(\mathbf{x}) = \sum_{i=1}^n x_i.$$

$$w(\mu) = \frac{\mu}{\sigma^2}$$

$$c(\mu) = \exp\left(-\frac{n\mu^2}{2\sigma^2}\right)$$

$$h(\mathbf{x}) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2\right)$$

By the Factorization Theorem,

$\sum X_i \text{ is a sufficient statistic for } \mu.$

Or equivalently, $\bar{X}$ is sufficient for μ , since it is a one-to-one transformation of $\sum X_i$.

Case 2: μ and σ^2 both unknown

Let

$$X_1, \dots, X_n \sim N(\mu, \sigma^2), \quad \mu \in \mathbb{R}, \sigma^2 > 0.$$

The joint density is

$$f(\mathbf{x} \mid \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right).$$

Expanding the quadratic,

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2.$$

Substituting,

$$f(\mathbf{x} \mid \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2 + \frac{\mu}{\sigma^2} \sum_{i=1}^n x_i - \frac{n\mu^2}{2\sigma^2}\right).$$

This has the Casella–Berger exponential-family form

$$f(\mathbf{x} \mid \mu, \sigma^2) = h(\mathbf{x}) c(\mu, \sigma^2) \exp\left(w_1(\mu, \sigma^2) \sum_{i=1}^n x_i + w_2(\mu, \sigma^2) \sum_{i=1}^n x_i^2\right),$$

with

$$t_1(x) = x, \quad t_2(x) = x^2.$$

Therefore, by the exponential-family sufficiency theorem,

$$T(\mathbf{X}) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right) \text{ is sufficient for } (\mu, \sigma^2).$$

Equivalently, the statistic

$$\left(\bar{X}, \sum_{i=1}^n (X_i - \bar{X})^2 \right)$$

is sufficient, since it is a one-to-one transformation of $(\sum X_i, \sum X_i^2)$.

When both μ and σ^2 are unknown, the Normal distribution is no longer a one-parameter exponential family, so the conical form is no longer adequate. The Casella–Berger form allows multiple sufficient components, which is why we use it here.

Minimal Sufficient Statistics

Definition 2: A statistic T is *minimal sufficient* if (i) it is sufficient and (ii) if S is any sufficient statistic, then T is a function of S .

If we use the book analogy again, recall that a book contains a secret – the whole book is $\mathbf{X}$ and θ is a secret message hidden in the book. Remember $T(\mathbf{x})$ is one paragraph that summarizes the secret message (i.e. θ), so T is a sufficient statistic. Further suppose that really only two sentences reveal the secret message, the other sentences just add flavor, context or repeated clues but *no new information*. The two sentences would be called a **minimal sufficient statistic**.

Example: We have shown that if $X_1, X_2, \dots, X_n \sim N(\mu, \sigma)$ then, $T(\mathbf{X}) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$ is sufficient for the parameter vector (μ, σ^2) .

But, there are other statistics that would also be sufficient.

Suppose $T_1(\mathbf{X}) = (X_1, X_2, \dots, X_n)$. If we had all of the data, then this is also sufficient because from the data we can calculate $\sum_{i=1}^n X_i$ and $\sum_{i=1}^n X_i^2$, so $T_1(\mathbf{X})$ is also sufficient, but it contains more information than $T(\mathbf{X})$.

Using the same reasoning, $T_2(\mathbf{X}) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2, X_3 \right)$ is also sufficient, but it too contains extraneous information.

Thus, $T(\mathbf{X}) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$ would be minimal sufficient because:

- every other sufficient statistic must contain enough information to compute these two quantities;
- but once you know these two values, **no smaller statistic** contains all the information about (μ, σ) that the full data contain. In other words, $T(\mathbf{X})$ is the smallest summary that retains 100% of the information about the parameters.

Likelihood Ratio Criterion for Minimal Sufficiency

Suppose we observe two datasets, $\mathbf{x}$ and $\mathbf{y}$, drawn from the same joint distribution with parameter θ . Different datasets may provide different amounts of evidence about plausible values of θ .

Now suppose that for *every* possible value of θ , the *relative plausibility* of observing $\mathbf{x}$ versus $\mathbf{y}$ is the same. That is, the ratio

$$\frac{f_{\mathbf{X}}(\mathbf{x} | \theta)}{f_{\mathbf{X}}(\mathbf{y} | \theta)}$$

does not depend on θ .

This means that, regardless of the true value of the parameter, $\mathbf{x}$ and $\mathbf{y}$ distinguish between possible values of θ in exactly the same way. From the model's point of view, they contain the same information about θ .

If two datasets always provide the same relative information about the parameter, then any sufficient statistic should treat them as equivalent.

Note: You've already seen this object before. When we view the joint density as a function of θ with the data fixed, we call it the likelihood. The criterion we used for minimal sufficiency is therefore called a likelihood-ratio criterion.

Theorem: (READING ONLY) Let $f(\mathbf{x}|\theta)$ be a joint pdf or pmf. The statistic $T(\mathbf{X})$ is minimal sufficient iff the following holds:

For any two samples $\mathbf{x}$ and $\mathbf{y}$.

$$\frac{f(\mathbf{x}|\theta)}{f(\mathbf{y}|\theta)} \text{ is constant in } \theta \iff T(\mathbf{x}) = T(\mathbf{y}) \quad (1)$$

This theorem says that any two data sets produce the same value of the minimal sufficient statistic, iff their likelihood ratio does not depend on the parameter. Sometimes this theorem is called the **Likelihood–Ratio Criterion** for Minimal Sufficiency.

In case you don't remember, an equivalence relation is defined as: $x \sim y \iff$ the ratio is constant in θ . We will use this idea to prove the theorem.

We prove both directions of the equivalence.

(1) ($\Rightarrow$) If T is sufficient, then the ratio condition holds.

Assume T is sufficient and suppose $T(\mathbf{x}) = T(\mathbf{y}) = t$.

Because T is sufficient, the conditional distribution $f_{\mathbf{X}|T}(\cdot | t, \theta)$ does not depend on θ . Therefore, the ratio

$$\frac{f_{\mathbf{X}|T}(\mathbf{x} | t, \theta)}{f_{\mathbf{X}|T}(\mathbf{y} | t, \theta)}$$

is constant in θ .

Using the conditional density identity,

$$f_{\mathbf{X}|T}(\mathbf{x} | t, \theta) = \frac{f_{\mathbf{X}}(\mathbf{x} | \theta)}{f_T(t | \theta)}, \quad T(\mathbf{x}) = t,$$

and similarly for $\mathbf{y}$, we obtain

$$\frac{\frac{f_{\mathbf{X}}(\mathbf{x} | \theta)}{f_T(t | \theta)}}{\frac{f_{\mathbf{X}}(\mathbf{y} | \theta)}{f_T(t | \theta)}} = \frac{f_{\mathbf{X}}(\mathbf{x} | \theta)}{f_{\mathbf{X}}(\mathbf{y} | \theta)}.$$

Therefore,

$$\frac{f_{\mathbf{X}|T}(\mathbf{x} | t)}{f_{\mathbf{X}|T}(\mathbf{y} | t)} = \frac{f(\mathbf{x} | \theta)}{f(\mathbf{y} | \theta)}$$

because the left hand side does not depend on θ (by sufficiency of T), the right hand side is also independent of θ . Hence,

$$\frac{f(\mathbf{x} | \theta)}{f(\mathbf{y} | \theta)}$$

is constant in θ .

(2) ($\Leftarrow$) If the ratio condition holds, then T is minimal sufficient.

Assume the likelihood–ratio condition:

$$\frac{f(\mathbf{x} | \theta)}{f(\mathbf{y} | \theta)} \text{ is constant in } \theta \iff T(\mathbf{x}) = T(\mathbf{y}). \quad (1)$$

We first show that T is sufficient.

Fix t and consider the set $\{\mathbf{x} : T(\mathbf{x}) = t\}$. Choose a reference point $\mathbf{x}_0$ such that $T(\mathbf{x}_0) = t$. For any $\mathbf{x}$ with $T(\mathbf{x}) = t$, condition (1) implies

$$\frac{f(\mathbf{x} | \theta)}{f(\mathbf{x}_0 | \theta)} = c(\mathbf{x}),$$

where $c(\mathbf{x})$ depends on $\mathbf{x}$ but not on θ . Hence

$$f(\mathbf{x} | \theta) = f(\mathbf{x}_0 | \theta) c(\mathbf{x}), \quad \text{for all } \mathbf{x} \text{ with } T(\mathbf{x}) = t.$$

Summing over all $\mathbf{y}$ such that $T(\mathbf{y}) = t$ yields

$$\sum_{\mathbf{y}: T(\mathbf{y})=t} f(\mathbf{y} | \theta) = f(\mathbf{x}_0 | \theta) \sum_{\mathbf{y}: T(\mathbf{y})=t} c(\mathbf{y}).$$

Therefore, the conditional density of $\mathbf{X}$ given $T = t$ is

$$\begin{aligned} f(\mathbf{x} \mid T = t, \theta) &= \frac{f(\mathbf{x} \mid \theta)}{\sum_{\mathbf{y}: T(\mathbf{y})=t} f(\mathbf{y} \mid \theta)} \\ &= \frac{f(\mathbf{x}_0 \mid \theta) c(\mathbf{x})}{f(\mathbf{x}_0 \mid \theta) \sum_{\mathbf{y}: T(\mathbf{y})=t} c(\mathbf{y})} \\ &= \frac{c(\mathbf{x})}{\sum_{\mathbf{y}: T(\mathbf{y})=t} c(\mathbf{y})}, \end{aligned}$$

which does not depend on θ . (For continuous models, the argument is identical with sums replaced by integrals.)

Thus T is sufficient.

To show minimality, let S be any other sufficient statistic. If $S(\mathbf{x}) = S(\mathbf{y})$, then sufficiency of S implies that

$$\frac{f(\mathbf{x} \mid \theta)}{f(\mathbf{y} \mid \theta)}$$

is constant in θ . By condition (1), this implies $T(\mathbf{x}) = T(\mathbf{y})$. Hence T is a function of every sufficient statistic and is therefore minimal sufficient.

Minimal sufficiency partitions the sample space into the finest equivalence classes that still preserve all information about the parameter θ .

Example: Let $X_1, \dots, X_n$ be iid $N(\mu, \sigma^2)$ with σ^2 known.

The joint density is

$$f(\mathbf{x}|\mu) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right), \quad \mathbf{x} = (x_1, \dots, x_n).$$

We apply the likelihood-ratio criterion of Casella–Berger: for any two samples $\mathbf{x}$ and $\mathbf{y}$, examine

$$R(\mu) = \frac{f(\mathbf{x}|\mu)}{f(\mathbf{y}|\mu)}.$$

Compute the exponent in the ratio. Using the identity

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2,$$

(and the analogous expression for $\mathbf{y}$), we obtain

$$\begin{aligned} \log R(\mu) &= -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 + \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 \\ &= -\frac{1}{2\sigma^2} \left(\sum x_i^2 - 2\mu \sum x_i + n\mu^2 \right) + \frac{1}{2\sigma^2} \left(\sum y_i^2 - 2\mu \sum y_i + n\mu^2 \right) \\ &= -\frac{1}{2\sigma^2} \sum x_i^2 + \frac{1}{2\sigma^2} \sum y_i^2 + \frac{\mu}{\sigma^2} \left(\sum x_i - \sum y_i \right), \end{aligned}$$

since the $n\mu^2$ terms cancel. Hence

$$R(\mu) = \exp\left\{-\frac{1}{2\sigma^2} \sum x_i^2 + \frac{1}{2\sigma^2} \sum y_i^2\right\} \exp\left\{\frac{\mu}{\sigma^2} \left(\sum x_i - \sum y_i\right)\right\}.$$

The only dependence of $R(\mu)$ on μ occurs through the factor $\exp\left\{\frac{\mu}{\sigma^2} (\sum x_i - \sum y_i)\right\}$. Therefore $R(\mu)$ is a constant function of μ if and only if

$$\sum_{i=1}^n x_i - \sum_{i=1}^n y_i = 0, \quad \text{i.e.} \quad \sum_{i=1}^n x_i = \sum_{i=1}^n y_i.$$

By the previous theorem, we conclude that

$$\frac{f(\mathbf{x}|\mu)}{f(\mathbf{y}|\mu)} \text{ is constant in } \mu \iff \sum_{i=1}^n x_i = \sum_{i=1}^n y_i.$$

Thus the equivalence classes determined by the likelihood-ratio condition are exactly the level sets of the statistic $T(\mathbf{x}) = \sum_{i=1}^n x_i$.

Consequently $T(\mathbf{X}) = \sum_{i=1}^n X_i$ is a **minimal sufficient** statistic for μ (equivalently, $\bar{X}$ is minimal sufficient).

Note: In Casella/Berger you can find the example where σ^2 is not known.

Remark: If one prefers the statistic in normalized form, note that $\bar{X} = T/n$ is a one-to-one function of T and hence $\bar{X}$ is also minimal sufficient.

The point of the following theorem is that minimal statistics are essentially unique.

If two statistics are both:

- Sufficient - they keep all information about θ
- Minimal - they throw away all redundancy

then they cannot be fundamentally different objects. One is just a **relabeling** of the other. A 1-1 transformation changes the form without changing the information.

Minimal sufficient statistics are unique up to relabeling: if two statistics are both minimally sufficient, they encode exactly the same information about the parameter.

Theorem: (READING ONLY) Let $T_1(\mathbf{X})$ and $T_2(\mathbf{X})$ be two minimal sufficient statistics for the same family $\{f(\mathbf{x}|\theta) : \theta \in \Theta\}$. Then there exists a one-to-one (injective, i.e. 1-1) function g defined on the range of T_1 such that

$$T_2(\mathbf{X}) = g(T_1(\mathbf{X})) \quad (\text{a.e.}).$$

Note: a.e. means the equality may fail on a set of sample points that has probability 0, but that failure is irrelevant for probability and inference.

Equivalently, any two minimal sufficient statistics are the same up to a one-to-one transformation.

Proof: Because T_1 is minimal sufficient and T_2 is sufficient, minimality of T_1 implies that T_1 is a (measurable) function of T_2 . Thus there exists a measurable map h on the range of T_2 with

$$T_1(\mathbf{X}) = h(T_2(\mathbf{X})) \quad (\text{a.e.}) \quad (1)$$

Conversely, because T_2 is minimal sufficient and T_1 is sufficient, minimality of T_2 implies that T_2 is a function of T_1 . Thus there exists a measurable map g on the range of T_1 with

$$T_2(\mathbf{X}) = g(T_1(\mathbf{X})) \quad (\text{a.e.}) \quad (2)$$

Combining (1) and (2) gives, for almost every sample point,

$$T_1(\mathbf{X}) = h(T_2(\mathbf{X})) = h(g(T_1(\mathbf{X}))).$$

Hence the composition $h \circ g$ acts as the identity on the range of T_1 (almost everywhere):

$$(h \circ g)(t) = t \quad \text{for almost every } t \in \text{Range}(T_1). \quad (3)$$

Similarly, composing the other way yields

$$(g \circ h)(s) = s \quad \text{for almost every } s \in \text{Range}(T_2). \quad (4)$$

From (3) we deduce that g is injective on $\text{Range}(T_1)$: if $g(t) = g(t')$ then applying h to both sides gives

$$t = (h \circ g)(t) = (h \circ g)(t') = t',$$

so $t = t'$. Thus g is one-to-one (modulo null sets). Likewise h is the (almost-everywhere) inverse of g on the image of g , so g is a bijection (1-1 and onto) between $\text{Range}(T_1)$ and $\text{Range}(T_2)$ (up to sets of probability zero).

Therefore there exists a one-to-one mapping g (defined on the range of T_1) such that $T_2(\mathbf{X}) = g(T_1(\mathbf{X}))$ almost surely. This proves that any two minimal sufficient statistics are equivalent up to a one-to-one transformation.

The proof shows two things: (i) **every minimal sufficient statistic is a function of any other sufficient statistic**, and (ii) **if two statistics are both minimal sufficient they must be invertible transforms of each other (almost everywhere)**. In practice one usually ignores the *almost everywhere (a.e.)* technicality and simply says *unique up to a one-to-one transformation*.

The following example illustrates that if two statistics are both minimal sufficient for a parameter, then they must be one-to-one transformations of each other.

Example: Assume $X_1, X_2, \dots, X_n$ are iid from $N(\mu, 1)$. From the likelihood-ratio criterion we know that

$$T_1(\mathbf{X}) = \sum_{i=1}^n X_i$$

is minimal sufficient for μ .

What about $T_2(\mathbf{X}) = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$?

$T_2(\mathbf{X})$ is also (by factorization) minimal sufficient but $T_1(\mathbf{X})$ and $T_2(\mathbf{X})$ differ by a 1-1 transformation.

$$T_2(\mathbf{X}) = g(T_1(\mathbf{X})) = \frac{1}{n} T_1(\mathbf{X})$$

Where $g(t) = t/n$ which is clearly a 1-1 function.

Example: Assume $X_1, X_2, \dots, X_n$ are iid from Bernoulli(p).

$$T_1(\mathbf{X}) = \sum_{i=1}^n X_i$$

$$T_2(\mathbf{X}) = \frac{1}{n} \sum_{i=1}^n X_i$$

Note that $T_2(\mathbf{X}) = \frac{1}{n} T_1(\mathbf{X})$ which is a 1-1 transformation.

Summary

Distribution	Parameter(s)	Minimal sufficient statistic
Bernoulli(p)	p	$\sum X_i$
Poisson(λ)	λ	$\sum X_i$
Normal(μ, σ^2 known)	μ	$\sum X_i$ or $\bar{X}$
Normal(μ, σ^2 unknown)	μ, σ^2	$(\sum X_i), \sum X_i^2$
Exponential(λ)	λ	$\sum X_i$
Discrete Uniform($1, 2, \dots, \theta$)	θ	Max X_i